

Homework 3

MATH 518 - Fall 2025

Due on Sunday, November 2, 11:59pm on Crowdmark

Total of points = 40 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

Recall that we can embed the affine space in the projective space by the maps

$$\varphi_j: k^n \longrightarrow U_j := \{[x_1 : \cdots : x_{n+1}] \mid x_j \neq 0\} \subset \mathbb{P}^n(k),$$

for $1 \leq j \leq n+1$. The complement $H_\infty = \mathbb{P}^n(k) \setminus U_{n+1}$ is called the hyperplane at infinity. Given an affine algebraic set $V \subset k^n$, we defined the projective closure $V^ \subset \mathbb{P}^n(k)$ by $V^* = V(I(V)^*)$. It is a projective algebraic set that satisfies $\varphi_{n+1}(V) = V^* \cap U_{n+1}$. Conversely, given $W \subset \mathbb{P}^n(k)$, we define the affine algebraic set $W_* = V(I(W)_*) = W \cap U_{n+1}$. We have*

- $(V^*)_* = V$ for any affine algebraic set $V \subset k^n$,
- $(W_*)^* = W$ for any projective algebraic set $W \subset \mathbb{P}^n(k)$ such that W is not contained in H_∞ nor contains H_∞ .

From this and the Exercise 3 below, it follows that the maps $V \longmapsto V^$ and $W \longmapsto W_*$ induce a one-to-one correspondence between projective varieties $W \subset \mathbb{P}^n(k)$ that are not contained in H_∞ , and the affine varieties in k^n . You can use this in Exercise 4.*

1. (10 points) Let $H_1, \dots, H_m$ be hyperplanes in $\mathbb{P}^n(k)$. If $m \leq n$, show that

$$H_1 \cap \cdots \cap H_m \neq \emptyset.$$

2. (10 points) Let $V = V(f_1, f_2) \subset k^3$, where $f_1, f_2 \in k[x, y, z]$ are the polynomials given by $f_1 = y - x^2$ and $f_2 = z - x^3$. Let $I(V)^* \subset k[x, y, z, w]$ be the homogenization of the ideal $I(V) = (f_1, f_2)$. Show that $zw - xy \in I(V)^*$ but $zw - xy \notin (f_1^*, f_2^*)$.

We always have $(f_1^, \dots, f_r^*) \subset (f_1, \dots, f_r)^*$, but this example shows that we don't always have equality. However, for principal ideals we have seen in class that $(f)^* = (f^*)$.*

3. (10 points) Let $V \subset W \subset \mathbb{P}^n(k)$ be two (non-empty) projective varieties, where $V = V(f)$ is a hypersurface defined by a homogeneous irreducible polynomial f . Show that $W = V$ or $W = \mathbb{P}^n(k)$. Deduce that the projective varieties $W \subset \mathbb{P}^n(k)$ that contain H_∞ are $W = \mathbb{P}^n(k)$ and $W = H_\infty$. Determine W_* and $(W_*)^*$ in the two cases.
4. (10 points) Describe all projective varieties in $\mathbb{P}^1(k)$ and $\mathbb{P}^2(k)$.